

Sai Manikanta Karnati

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years building high-performance, cloud-native software platforms across full-stack and microservices for digital health and benefits domains like PatientPoint. Skilled in Python, TypeScript, JavaScript, React, Angular, AWS, Kubernetes, Docker, Elasticsearch, and SQL. Delivered HIPAA-aligned digital services and DevOps automation, improving reliability to 99.99% and cutting issue resolution time by 40%.

SKILLS

Python - TypeScript - JavaScript - React - Angular - AWS - Kubernetes - Docker - Microservices - DevOps - SQL - Elasticsearch - Agile - HIPAA - C - R - UI - Node.js - Spring Boot - GraphQL - Apache Kafka - Terraform - CI/CD - Technical Writing

EXPERIENCE

Software Engineer - Vanguard

2025 - present

- Architected scalable backend microservices in Python and Spring Boot, enabling independent deployments and reducing release lead time by ~30%.
- Built Python data processing pipelines and DevOps automation scripts, reducing manual operational effort by ~60% across 250+ workdays per year.
- Optimized complex SQL queries, stored procedures, and data models across relational and NoSQL databases, improving reporting performance by 35%.
- Engineered event-driven systems with DynamoDB Streams, Python/Node.js Lambda, and SNS/SQS, ensuring 99.99% reliability at high volume.
- Developed responsive Angular and React single-page UIs in TypeScript and JavaScript, lifting UI render performance by ~25% for digital platforms.
- Designed RESTful and GraphQL APIs delivering HIPAA-aligned patient, benefits, and compensation data securely to ~10K+ daily users.
- Deployed cloud-native apps on AWS (Lambda, ECS, EC2, S3, RDS, DynamoDB) with Docker and Kubernetes, integrating Elasticsearch for sub-second search.
- Implemented monitoring with Splunk and Spring Boot Actuator, reducing production issue resolution time by 40% and keeping uptime true to SLA.

Associate Software Engineer - Xebia

2022 - 2023

- Developed backend services in Python, Spring Boot, and Node.js, delivering scalable RESTful APIs and microservices for high-performance platforms.
- Built Python automation scripts for data transformation and validation, cutting manual effort by ~50% across recurring batch jobs.
- Designed and optimized SQL queries, joins, views, and stored procedures, improving API response times by 35%.
- Built dynamic Angular and React UI frontends in TypeScript and JavaScript with RxJS and Material, improving usability for ~5K users.
- Authored technical writing and Swagger/OpenAPI docs, making APIs easy to read and reducing onboarding time for new engineers by ~30%.
- Implemented event-driven architecture with Apache Kafka and Elasticsearch indexing, reducing inter-service failures by 35%.
- Automated infrastructure with Terraform on AWS, applying DevOps and Agile practices to ship faster across release cycles.
- Containerized applications with Docker and Kubernetes, improving CI/CD deployment efficiency by ~30%.

Engineer Trainee - Xebia

2021 - 2021

- Developed enterprise applications using Angular frontends and Python/Java backend services in an Agile team.

- Built and optimized SQL queries, stored procedures, and relational schemas, improving query execution time by 20%.
- Wrote reusable code in Python, C, and R for data validation and automation supporting backend pipelines.
- Developed RESTful APIs and documented service contracts with Swagger/OpenAPI for clear, readable interfaces.
- Built Angular UI modules in TypeScript with Redux and Reactive Forms for transactional digital platforms.
- Increased unit test coverage to 95% using JUnit and Python-based test automation tools.

EDUCATION

Master of Science in Computer Science - University Of North Carolina At Charlotte

CERTIFICATIONS

- AWS Certified Developer Associate
- MongoDB Certified Developer
- Infosys Certified Software Programmer
- NPTEL Certified Java Developer
- AWS Academy Graduate - AWS Academy Data Engineering